

Railtrack Company Code of Practice

METHODOLOGY FOR THE DEMONSTRATION OF COMPATIBILITY WITH INTERLOCKINGS

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ISSUE RECORD

ISSUE	DATE	COMMENTS
I	February 2003	New Code of Practice

IMPLEMENTATION

The provisions of this code of practice may be used from the date of issue.

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Published & Issued by **Railtrack PLC (part of the Network Rail group of companies)**
Railtrack House
Euston Square
LONDON
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Contents

1. INTRODUCTION.....	4
1.1 PURPOSE	4
1.2 SCOPE	4
2. SOURCE DOCUMENTS.....	4
2.1 PREVIOUS SAFETY CASE DOCUMENTATION	4
2.2 OTHER STANDARDS	5
2.3 OTHER DOCUMENTS	5
2.4 IDI DOCUMENTS.....	5
3. DESCRIPTION OF EQUIPMENT.....	6
3.1 ROUTE RELAY INTERLOCKING (RRI).....	6
3.2 SOLID STATE INTERLOCKING (SSI).....	6
3.2.1 <i>Control Centre Equipment</i>	7
3.2.2 <i>Trackside Functional Modules (TFM)</i>	7
3.2.3 <i>Data Links</i>	8
4. INFRASTRUCTURE ASSUMPTIONS	8
5. SUSCEPTIBILITY ANALYSIS.....	8
5.1 RADIATED, CONDUCTIVE AND INDUCTIVE INTERFERENCE-RELAY INTERLOCKING ..	9
5.2 LARGE RAIL-TO-EARTH VOLTAGES CONDUCTED INTO TFM	9
5.2.1 <i>Generation of Voltage Spikes</i>	9
5.2.2 <i>Capacitive Coupling</i>	10
5.2.3 <i>Magnitude of the Spikes</i>	10
5.2.4 <i>Effect of Spikes on SSI</i>	11
6. METHODOLOGY FOR DEMONSTRATION OF COMPATABILITY.....	11
6.1 RELAY INTERLOCKING CASE	11
6.2 SSI INTERLOCKING CASE	12

I. INTRODUCTION

I.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility with interlocking systems installed on the a.c. and d.c. electrified railway on Railtrack controlled infrastructure. This is achieved by specifying criteria/standards against which test results from new rolling stock should be compared to enable a safety case to be made.

I.2 Scope

This document covers:

- Route Relay Interlocking (RRI) systems installed on a.c. and d.c. electrified Railtrack controlled infrastructure. This only includes relay room installations. External line circuits, TDM systems and vital Reed FDM systems which may form part of the interface between the RRI and trackside equipment are covered by other IDI documents [14, 15, 16].
- Solid State Interlocking (SSI) systems installed on a.c. and d.c. electrified Railtrack controlled infrastructure. This includes control centre equipment and Trackside Functional Modules (TFMs) which are manufactured by Westinghouse and ALSTOM in accordance with Railtrack specifications. Data Links (TDLs) and Long Line Links (LLL) which are used to send SSI data are covered by another IDI document [16].

Note: In the context of this document, “Solid State Interlocking” refers to the system developed by the British Railways Research Division in the 1970s and completed under a tripartite agreement between BR, GEC General Signal Ltd and Westinghouse Signal Ltd. This document does not cover other types of computer based interlockings.

2. Source Documents

2.1 Previous Safety Case Documentation

1. “Juniper Project – Electric Traction Interference: Non Track Circuit Systems” AEA Technology Rail Ref. PRJ-SCSE(JUN)-RP-16, Issue 2, September 1998.
2. “Compatibility of Class 357 Electrical Systems with Miscellaneous Systems on LTS Routes”, ADtranz Ref. 3EER300000-1179, Issue D, July 2000.
3. “Class 365/5 Trains: AC Full Service EE&CS EMC Safety Case Assessment-Module 3 Other Signalling Systems”, AEA Technology Ref. PRJ-SCSE(ELAC)-SA-18, Issue 2,

April 1997.

4. "Class 92 Locomotive EMC safety Assessment Volume 8-Miscellaneous Signalling Equipment", AEA Technology Rail Ref. 120-dr0442, Issue b0, May 1999.
5. "Class 92 Locomotive EMC Safety Assessment Volume 11-Miscellaneous Equipment on 750 V DC Railway, AEA Technology Rail Ref. 120-dr0077, Issue d0, October 1997.
6. "Rail to Earth Voltages Compatibility for SSI - Juniper AC Train – Class 334- 2x3-Car Trains", ALSTOM's Ref. 10149.1238, Issue B, March 2000.

2.2 Other Standards

7. "Electromagnetic Susceptibility Data for Track Circuits and AWS", Group Standard GS/ES1914, Issue 1, BRB, Feb 1994. (It is understood that this document will shortly be withdrawn)
8. "Earthing and Bonding of SSI Equipment – SSI Applications Manual" SSI8503, Issue 1, June 1999.
9. "Railway Applications Electromagnetic Compatibility- ENV 50121 Part 3-1", European EMC Standard for Railways.
10. "Design of SSI Schemes – SSI Application Manual", SSI8500A", Issue 6, August 2000.
11. "Installation of Signalling and Operational Telecommunications Equipment" Railway Group Standard GK/RT0208 issue 1, August 2000.
12. "Signalling Installation" Railtrack Line Code of Practice RT/E/C/11210, issue 1, August 2000

2.3 Other Documents

13. "Details on Design, Development and Application of Solid State Interlocking on British Rail", IRSE SSI Green Booklet, No. 29, D H Stratton, First Edition, 1988.

2.4 IDI Documents

14. RT/E/C/50014 Methodology for the demonstration of compatibility with Lineside Equipment on AC and DC Railways.
15. RT/E/C/50015 Methodology for the demonstration of compatibility with Reed FDM

16. RT/E/C/50016 Methodology for the demonstration of compatibility with Telecomms Systems.

3. Description of Equipment

The railway signalling interlocking is a collection of mechanisms (mechanical, electro-mechanical, relay or solid state) used to control the operation of signals and points, in order that a train may pass from one section of line to the next in complete safety. The interlocking will prevent conflicting routes being set up by the signaller or by the Automatic Route Setting (ARS) equipment. The principles of interlocking remain the same whatever the mechanism used.

3.1 Route Relay Interlocking (RRI)

In route relay interlocking, the setting and releasing of routes and the safe control of signals, points and other items of trackside equipment is effected by the use of dc relay technology. The system consists either of discrete relays (free-wired interlocking) or ready-made standard units with additional discrete relays (geographical interlocking). The relays used are miniature d.c. tractive armature types which are known to be reliable and “fail-safe”; i.e., they are certain to release after de-energisation.

3.2 Solid State Interlocking (SSI)

Solid State Interlocking (SSI) is based on the use of programmable solid state electronics (microprocessors), in place of relays, to carry out the setting and releasing of routes and the safe control of signals, points and other items of trackside equipment. Unlike relays, the failure modes of electronic devices are essentially unpredictable so that redundancy techniques are used to ensure safe operation. Processes are carried out by separate, though identical modules. Each module compares its own result with those from other modules, and actions are initiated only if acceptable agreement is reached. Redundant modules operate with a controlled time displacement between the parallel processes; this greatly reduces the risk that severe external interference will cause different modules to be adversely affected in exactly the same way.

The interlocking is controlled by software, which interprets geographic data specific to a particular installation. Both software and data is subject to a rigorous validation process to control the risk of systematic errors. SSI consists of equipment in the control centre, at the trackside and data links connecting the two. The following sections give a brief description of SSI equipment covered by this report.

3.2.1 Control Centre Equipment

Each electronic interlocking consists of a rack of microprocessor based equipment housed in the control centre. This equipment performs the principal logical functions necessary for safe control of trains within a defined area. The number of interlockings required depends on the size and complexity of the area to be controlled from the control centre. Interlockings are required to have a very high availability, so the main functional elements (notably the interlocking processor modules - MPM) are triplicated and normally operate in a "two-out-of-three" majority voting configuration. The voting process isolates any interlocking processor module which is out of agreement with its partners, leaving the two remaining modules in a safe "two-out-of-two" configuration until the faulty module is replaced.

Information flow between the interlocking and the signalman's display system (which may be an entrance/exit panel, IECC or other VDU based system) is handled by a Panel Processor Module (PPM). The PPMs are duplicated in order to provide fault tolerance.

An installation using an entrance/exit panel requires a Panel Multiplexer. This generally consists of a microcomputer system which scans the controls on the panel, generating a serial data stream which it sends to the PPMs. Similarly, serial data from the PPMs is demultiplexed to drive the panel indications. A Panel Multiplexer may be duplicated to increase availability.

The control centre equipment also includes a Technician's Terminal enabling the operator to apply certain controls, obtain fault information and provides the facility of logging the state of the signalling system onto tape. The Technician's Terminal interfaces with the interlocking via a dedicated Diagnostic Module.

3.2.2 Trackside Functional Modules (TFM)

SSI controls signals and points directly by means of two types of TFMs: Signal Modules and Point Modules, which are situated in lineside location cases. Each TFM has a number of vital outputs for lighting signal lamps or operating point motors and a number of inputs for lamp current proving or point detection. Inputs are also available for sensing the state of external contacts (e.g., track circuit relays). These are fed with a coded signal which is verified by the module to provide protection against false feeds. The TFMs contain two microprocessor modules operating in a duplicated configuration.

If the module detects that it is faulty or loses communication with the interlocking, it will automatically switch off all other outputs and connect the Aspect Retaining feed which is wired to the red or most restrictive aspect of the signal. This ensures that a signal will display its most restrictive aspect in the event of failure.

3.2.3 Data Links

Interlockings are generally connected to TFM's by means of duplicated trackside data links. These consist of individually screened twisted pair copper cables having a nominal impedance of 100 ohms to specification BR 1932, the equipment at each location being connected in parallel via Data Link Modules (DLMs). Similar data links are used to connect two or more interlockings allowing vital data to be exchanged between them. SSI messages incorporate encoding and other features that reduce their susceptibility to traction interference. For distances in excess of 40km, 64kbit/s voice channels in "open" 2Mbit/s PCM telecommunications circuits are used. These are interfaced to the SSI equipment by means of Long Distance Terminals. These incorporate redundant processors and provide buffering and further levels of encoding of SSI messages.

A full discussion of compatibility issues between traction systems and SSI data transmission equipment is contained in the IDI Lineside Equipment document [14].

4. Infrastructure Assumptions

The methodology presented utilises the following assumptions:

- Interlocking systems have been installed and maintained in accordance with Group Standards or standards applicable at the time of installation and manufacturer's recommendations where applicable. In particular, systems have been installed in accordance with the principles defined in the Group Standard "Installation of Signalling and Operational Telecommunications Equipment" [11] and to the standards laid down in the "Signalling Installation" code of practice [12], especially with regard to power supplies, earthing and electronic equipment. More specifically, SSI schemes have been designed and installed in accordance with the requirements of the SSI Application Manual SSI8500 "Design of SSI Schemes" [10] and "Earthing and Bonding of SSI Equipment" [8].
- The interlocking systems as they currently exist, operate to an acceptable level of safety when used with the existing traction and rolling stock.

5. Susceptibility Analysis

The mechanisms whereby interference from electric traction systems can enter RRI and SSI interlocking systems are:

5.1 Radiated, Conductive and Inductive Interference-Relay Interlocking

The relays and circuits are installed in a building with effective earthing and bonding provided as detailed in Railtrack Signalling Design Handbook. The circuit lengths within the interlocking are small and hence not susceptible to interference. The risks associated with the external cabling are covered by another IDI document [16].

5.2 Large Rail-to-Earth Voltages conducted into TFM

The mechanism whereby pantograph related voltage spikes are generated and the effects on TFMs are discussed in details in this section.

5.2.1 Generation of Voltage Spikes

A spark is formed when the pantograph head and the overhead line (OHL) are not in physical contact and the air gap between them breaks down under the intense electric field. When this happens, the pantograph voltage rises or falls rapidly towards the voltage of the OHL; when the voltage across the air gap is no longer sufficient to support conduction, the spark is extinguished. Sparking is likely to re-occur during each subsequent half cycle of the supply variation. This repetitive sparking process ceases when movement of the pantograph and/or movement of the OHL makes the air gap too large to be broken down, or when the pantograph head and the OHL make physical contact.

Sparking can be expected to occur when a pantograph is being raised or lowered, and when the train is entering or leaving a neutral section. In these cases, the main circuit breaker is open, and the only electrical load on the pantograph is the capacitance and the leakage resistance between the pantograph and the vehicle body. Unless any electrical components are installed which are not isolated by the main circuit breaker, these have negligible loading effect on the OHL supply. For practical purposes, the change in the pantograph voltage may be regarded as a step function.

A more complex case exists when the circuit breaker is closed and a momentary loss of contact occurs during train movement. In this situation, the electrical load on the pantograph includes the main transformer and the associated traction and auxiliary equipment. Such a load is inductive and loss of contact causes a low voltage arc to be drawn in an attempt to maintain any current which is flowing. In this case, the step change in voltage does not occur. Moreover, if the current is large enough, the OHL and pantograph head develop "hot spots" which allow the arc to re-strike each successive half cycle, effectively maintaining continuous conduction between the OHL and the pantograph. The arc is extinguished only if the current is too small to maintain cathodic emission by the hot spots, or if movement of the train draws the arc to an excessive length. Thus, sparking is limited to the period between any arc

being extinguished and the pantograph again making contact with the OHL.

5.2.2 Capacitive Coupling

When a spark occurs there is an abrupt change in the pantograph voltage. This voltage step, which may be positive or negative going, is applied to the High Tension (HT) circuits of the traction unit. In particular, it is applied to a potential divider formed by the series connection of:

- the capacitance between the pantograph and the vehicle body which includes the effect of transformer and cable capacitances, and
- the impedance between the rails and earth.

The result is a positive or negative voltage spike which appears between the rails and earth. The magnitude and duration of the spike both increase with increasing pantograph-body capacitance and with increasing rail-earth impedance.

5.2.3 Magnitude of the Spikes

The pantograph-body capacitance is normally determined by the stray capacitance of the roof bushing, the effect of transformer and cable capacitances, earth switches, and the main circuit breaker. The construction of the traction transformer inevitably provides a capacitance between the winding and the core and other earthed metal of the tank. Different capacitance values are listed in Table I below for several traction transformers [6].

Calculated Values Class of Vehicle	Parallel Capacitance C_p from HT Line Terminal to Transformer Tank (pF)	Series Capacitance C_s between HT Terminals (pF)
Class 334 - Juniper	3200	1050 (layer winding)
Class 90 – Locomotive	3300	100 (disc winding)
Class 319, 325 & 365/5 EMU	1900	3 (disc winding)
CLASS 373 -EUROSTAR	8000	1400 (LAYER WINDING)

The value of the pantograph body capacitance is fixed for a given traction unit design,

and is reasonably stable and consistent throughout the vehicle life.

The rail-earth impedance is principally determined by the rail-earth leakage resistance, but is also affected by the characteristics of the traction return system. The magnitude of the leakage resistance varies considerably from one location to another and with changes in weather conditions. This variation makes it difficult to predict the magnitude and duration of the voltage spikes.

5.2.4 Effect of Spikes on SSI

The voltage spikes appearing on the rails are transferred directly to SSI equipment via track circuit tail cables. Although the track circuit relays notionally provide electrical isolation (typically a 1 kV ac dielectric test voltage is specified for Br933A, BR938A and BR966 relays), these voltage spikes have very steep leading edges and can be transferred into TFM's directly through electrical breakdown within the relays, or indirectly via parasitic capacitance or earth connections.

If the voltage spike is large enough to penetrate the protection provided by the input/output and power supply circuits of a TFM, it is likely to disrupt the operation of two microprocessors which work together in a "two-out-of-two" redundant configuration, providing protection against random component failures. The processors are deliberately operated out of synchronism so as to minimise the risk of common cause failures affecting both processors in exactly the same way. Thus, the consequence of any such disruption is a divergence between the two processors; this results in the blowing of the TFM security fuse.

Mark II TFM's have experienced these kinds of interference problems in the past. This was due to high values of pantograph to vehicle capacitance (of the order of 14000 pF for class 312) where capacitor dividers were fitted to EMUs for 6.25 kV/25 kV detection (now obsolete).

6. Methodology for Demonstration of Compatibility

6.1 Relay Interlocking Case

As stated in Section 7.1, traction interference, which might affect the operation of the RRI system, is mainly associated with the external line circuits. The RRI installations in the relay room can therefore be regarded as compatible with the traction interference generated by electric trains. However, when considering compatibility with the whole RRI system, the risk from traction interference to the external line circuits of the RRI system should be carefully assessed [15, 16].

6.2 SSI Interlocking Case

To prevent maloperation of SSI TFMs, a maximum rail/earth voltage of 1 kV peak was specified to limit pantograph related rail voltage transients (*This limit is quoted from Group Standard GS/ES1914 which is not referenced here, as it is likely to be withdrawn shortly*). However, as discussed in Section 7.2.3, the magnitude of this transient depends upon rail/earth impedance (a property which varies considerably depending on the traction return systems and weather dependent characteristics of the track). Thus, it is more satisfactory to demonstrate that pantograph-to-body capacitance is no greater than that on existing traction units (Table 1 gives some figures for the existing trains).

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019

Issue 1

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

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